A NEW REPRESENTATION OF THE FUNCTION PSI IN TERMS OF THE BASIC FERMIONIC OPERATORS

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Allowed standard prerequisites (editorial).

- The setup of the fermionic basis and the known integral representation of Psi_nu. Same paper, Section 2 (Review of the previous results), in force in the statement. Fixes the operators, the kernel F-bar and the equation the proof compares against.
- The epsilon-expansion of the functions zeta_0, zeta_R and of the kernel F-bar. Same paper, the expansions displayed at proof lines 723-740. Supplies the order-by-order comparison used in the induction.
- Residue calculus for the fermionic operators. Same paper, Section 2, the residue conventions. Underlies every displayed computation in the appendix.

$$\Psi(l,\kappa)+\frac{\mathrm{i}f\kappa}{l\big(l+\frac{\mathrm{i}}{2\nu}\big)}=-\sum_{r=1}^{k}\operatorname{res}_h\left[\frac{\mathrm{e}^{-hx_r^+(\kappa)/f\kappa}-\mathrm{e}^{-hx_r^-(\kappa)/f\kappa}}{h(l+h)}S(h)\right]$$

(3.1)

$$-\frac{2\mathrm{i}}{f\kappa}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^{\infty}\frac{\mathrm{d}x}{2\pi}\left\{\operatorname{res}_h\left[\frac{\mathrm{e}^{-hx/f\kappa}}{l+h}S(h)\right]\bar{F}(x,\kappa)^n\left(-\frac{\partial}{\partial x}\right)^n\right\}_{\text{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big).$$

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$$\Psi_\nu(s,p) := \frac{1}{2\nu\mathrm{i}}\Psi\Big(\frac{s}{2\nu\mathrm{i}},p\Big), \qquad S_\nu(h) := S\Big(\frac{h}{2\nu\mathrm{i}}\Big). \tag{3.2}$$

(3.3)

$$\begin{aligned}
&\zeta_0(s,p) = \zeta_0(s,p),\\
&\zeta_1(s,p) = \operatorname{res}_h\big[\sigma_1(h)\zeta_0(h,p)p^{-1}(1+s-h)\zeta_0(1+s-h,p)\big],\\
&\zeta_2(s,p) = \operatorname{res}_h\big[\big(\sigma_2(h)\zeta_0(h,p)+\sigma_1(h)\zeta_1(h,p)\big)p^{-1}(1+s-h)\zeta_0(1+s-h,p)\big]\\
&\qquad+\frac{1}{2}\operatorname{res}_h\operatorname{res}_{h'}\big[\sigma_1(h)\zeta_0(h,p)\sigma_1\big(h'\big)\zeta_0\big(h',p\big)p^{-2}\big(1+s-h-h'\big)_2\zeta_0\big(2+s-h-h',p\big)\big].
\end{aligned}$$

Proposition 3.1. *The function $\Psi_\nu(s,p)$ is given by the formula*

$$\Psi_\nu(s,p)|0\rangle = \exp\{\operatorname{res}_h[(S_\nu(h)-1)\Psi_\nu(h,p)\Phi_+(h)]\}\mathcal{F}(s,p,\Phi_-)|0\rangle, \tag{3.4}$$

where $\mathcal{F}(s,p,\Phi_-)$ is a functional of fields Φ_-,

$$\begin{aligned}
&\mathcal{F}(s,p,\Phi_-)\\
&\quad=\sum_{n=0}^{\infty}\frac{p^{-n}}{n!}\operatorname{res}_{h_1}\cdots\operatorname{res}_{h_n}\Phi_-(h_1)\cdots\Phi_-(h_n)\Big(1+s-\sum h_j\Big)_n\zeta_0\Big(n+s-\sum h_j,p\Big).
\end{aligned}$$

4. Proof of Proposition 3.1

In this appendix, we give the proof of Proposition 3.1. For simplicity, we consider the case without any particle-hole excitations. We want to show that the equation (3.4) gives the same result as the equation (3.1) in every order of the ϵ-expansion. In "renormalised" variables (3.2), the representation (3.1) of Ψ_ν takes the form

$$\begin{aligned}
&\Psi_\nu(s,p) - p\Psi_0(s,p)\\
&\quad=-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\left\{\operatorname{res}_h\left[\frac{\mathrm{e}^{\frac{\mathrm{i}hx}{p}}}{s+h}S(h)\right]\bar F(x,p)^n\left(-\frac{\partial}{\partial x}\right)^n\right\}_{\text{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big),
\end{aligned} \tag{4.1}$$

where the function $\bar F(x,p)$ is given by

$$\bar F(x,p) = x + \mathrm{i}\operatorname{res}_h\big[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}S(h)\Psi_\nu(h,p)\big]. \tag{4.2}$$

Let us introduce the expansion of the function $\bar F(x,p)$ with respect to ϵ,

$$\bar F(x,p) = \sum_{j=0}^{\infty}\bar F_j(x,p)\epsilon^j.$$

From (4.2), it follows that

$$\begin{aligned}
&\bar F_0(x,p) = x+\mathrm{i}\operatorname{res}_h\big[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}\zeta_0(h,p)\big] = x+\operatorname{res}_h\left[\mathrm{i}\mathrm{e}^{\frac{\mathrm{i}hx}{p}}\frac{p}{h(h-1)}\right] = \mathrm{i}p\sum_{m=2}^{\infty}\frac{1}{m!}\left(\frac{\mathrm{i}x}{p}\right)^m,\\
&\bar F_1(x,p) = \mathrm{i}\operatorname{res}_h\big[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}\sigma_1(h)\zeta_0(h,p)\big],\\
&\bar F_2(x,p) = \mathrm{i}\operatorname{res}_h\big[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}(\sigma_2(h)\zeta_0(h,p)+\sigma_1(h)\zeta_1(h,p))\big], \qquad \ldots.
\end{aligned} \tag{4.3}$$

The ϵ-expansion of the $\Psi(h,p)$ can be obtained from (4.1),

$$\begin{aligned}&\Psi_\nu(s,p)-p\Psi_0(s,p)\\&=-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\Big\{\operatorname{res}_h\Big[\frac{\mathrm{e}^{\frac{\mathrm{i}hx}{p}}}{s+h}\Big]\Big(F_0(x,p)^n\epsilon\cdot nF_0(x,p)^{n-1}F_1(x,p)\\&+\epsilon^2\Big(nF_0(x,p)^{n-1}F_2(x,p)+\frac{n(n-1)}{2}F_0(x,p)^{n-1}F_1(x,p)^2\Big)+\cdots\Big)\Big(-\frac{\partial}{\partial x}\Big)^n\Big\}_{\mathrm{even}}\\&\times\log\big(1+\mathrm{e}^{-2\pi x}\big).\end{aligned}\tag{4.4}$$

As it can be seen, in every order of the ϵ-expansion of the function $\Psi(s,p)$ one has to calculate the integral of the form

$$-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\Big\{\mathrm{e}^{-\frac{\mathrm{i}sx}{p}}\bar F_0(x,p)^n\Big(-\frac{\partial}{\partial x}\Big)^{n+n'}\Big\}_{\mathrm{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big)\tag{4.5}$$

for some $n'\geq 0$. To do so, let us introduce the following notation:

$$\frac{1}{n!}\Big[\sum_{k=2}^{\infty}\frac{x^k}{k!}\Big]^n=\sum_{m=2n}^{\infty}u_{n,m}x^m.$$

The numbers $u_{n,m}$ can be written down explicitly in the form

$$u_{n,m}=\frac{1}{n!}\sum_{\substack{2\leq k_1\leq\cdots\leq k_n\leq m-2(n-1),\\ k_1+\cdots+k_n=m}}\frac{1}{k_1!\cdots k_n!}.$$

The most important property of $u_{n,m}$ relevant to this proof is the following sum:

$$\sum_{\substack{0\leq n\leq r,\\ 0\leq k\leq n+r}}(-1)^{n+k}\frac{(n+r)!}{k!}u_{n,r+n-k}s^k=(-1)^r(s+1)_r.\tag{4.6}$$

Then the left-hand side of the formula (4.5) can be written as

$$\begin{aligned}&-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\Big\{\mathrm{e}^{-\frac{\mathrm{i}sx}{p}}\bar F_0(x,p)^n\Big(-\frac{\partial}{\partial x}\Big)^{n+n'}\Big\}_{\mathrm{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big)\\&=-\frac{2}{p}\sum_{\substack{n\geq 0,\\ k\geq 0,\\ m\geq 2n}}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\Big\{\mathrm{i}^{-k+n+m}p^{n-m-k}x^{k+m}\frac{s^k}{k!}u_{n,m}\Big(-\frac{\partial}{\partial x}\Big)^{n+n'}\Big\}_{\mathrm{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big)\\&=-2\sum_{\substack{n\geq 0,\\ k\geq 0,\\ m\geq 2n}}\mathrm{i}^{-k+n+m}p^{n-m-k-1}\frac{s^k}{k!}u_{n,m}(m+k)!\big(1-2^{-m-k-1+n+n'}\big)\\&\times\frac{\zeta\big(m+k-n-n'+2\big)}{(2\pi)^{m+k-n-n'+2}}.\end{aligned}\tag{4.7}$$

In the final line, the equality

$$\int_0^\infty\frac{\mathrm{d}x}{2\pi}x^m\Big(-\frac{\partial}{\partial x}\Big)^n\log\big(1+\mathrm{e}^{-2\pi x}\big)=m!\big(1-2^{-m-1+n}\big)\frac{\zeta(m-n+2)}{(2\pi)^{m-n+2}}$$

was used and it was assumed that $k+m-n-n'$ is even.

Let us denote $m+k-n=r$. Since $u_{n,m}=0$ for $m<2n$, the sums over k and n in (4.7) are actually finite. Using the property (4.6) of the numbers $u_{n,m}$, we continue the derivation

$$(4.7) = -2\sum_{\substack{n\geq 0,\\ k\geq 0,\\ r\geq 0}} p^{-r-1}(-1)^{n+k}\mathrm{i}^r\frac{(r+n)!}{k!}u_{n,r-k+n}s^k\big(1-2^{-r+n'-1}\big)\frac{\zeta\big(r-n'+2\big)}{(2\pi)^{r-n'+2}}$$

$$= -2\sum_{\substack{0\leq n\leq r,\\ 0\leq k\leq r-n,\\ r\geq 0}} p^{-r-1}(-1)^{n+k}\mathrm{i}^r\frac{(r+n)!}{k!}u_{n,r-k+n}s^k\big(1-2^{-r+n'-1}\big)\frac{\zeta\big(r-n'+2\big)}{(2\pi)^{r-n'+2}}$$

(4.8)

$$= -2\sum_{r\geq 0}(-\mathrm{i})^r p^{-r-1}(s+1)_r\big(1-2^{-r-n'+1}\big)\frac{\zeta\big(r-n'+2\big)}{(2\pi)^{r-n'+2}}.$$

We can denote $r-n'+2=2m$ since it is even. Then

$$(4.8) = -2\sum_{m=-n'+2}^{\infty}(-\mathrm{i})^{2m+n'-2}p^{-2m+1-n'}(s+1)_{2m+n'-2}\big(1-2^{-2m+1}\big)\frac{\zeta(2m)}{(2\pi)^{2m}}$$

$$= -2(\mathrm{i}p)^{-n'}(1+s)_{n'}\frac{1}{s+n'}\sum_{m=1}^{\infty}p^{-2m+1}\big(s+n'\big)_{2m-1}\big(1-2^{-2m+1}\big)(-\mathrm{i})^{2m-2}\frac{|B_{2m}|}{2}$$

$$- 2(\mathrm{i}p)^{-n'}\sum_{m=-n'+2}^{0}\mathrm{i}^{2m-2}p^{-2m+1}(1+s)_{2m+n'-2}\big(1-2^{-2m+1}\big)\frac{\zeta(2m)}{(2\pi)^{2m}}$$

$$= (\mathrm{i}p)^{-n'}(1+s)_{n'}\zeta_R\big(s+n',p\big) - \mathrm{i}(\mathrm{i}p)^{-n'+1}(1+s)_{n'-2}$$

$$- 2(\mathrm{i}p)^{-n'}\sum_{m=-n'+2}^{0}\mathrm{i}^{2m-2}p^{-2m+1}(1+s)_{2m+n'-2}\big(1-2^{-2m+1}\big)\frac{\zeta(2m)}{(2\pi)^{2m}}.$$

For $n'\geq 2$, the only term that contributes to the sum in the last line is the one with $m=0$. Otherwise the sum is zero. Therefore, we get to the following result:

$$-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^{\infty}\frac{\mathrm{d}x}{2\pi}\left\{\mathrm{e}^{-\frac{\mathrm{i}sx}{p}}\bar{F}_0(x,p)^n\left(-\frac{\partial}{\partial x}\right)^{n+n'}\right\}_{\text{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big)$$

$$(4.9)\qquad = \begin{cases}(\mathrm{i}p)^{-n'}(1+s)_{n'}\zeta_R\big(s+n',p\big) & \text{for } n'=0,1,\\ (\mathrm{i}p)^{-n'}(1+s)_{n'}\zeta_0\big(s+n',p\big) & \text{for } n'\geq 2.\end{cases}$$

From this result for $n'=0$, it follows immediately that the ϵ^0-term of the expansion (4.1) is $\zeta_0(s,p)$ which is in agreement with Proposition 3.1. Now let us turn our attention to the first order with respect to ϵ. From (4.4) and (4.3), we get

$$\zeta_1(s,p) = -\frac{2}{p}\left(\sum_{n=1}^{\infty}\frac{\mathrm{i}}{n!}\int_0^{\infty}\frac{\mathrm{d}x}{2\pi}\left\{\mathrm{e}^{-\frac{\mathrm{i}sx}{p}}n\bar{F}_0(x,p)^{n-1}\operatorname{res}_h\left[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}\sigma_1(h)\zeta_0(h,p)\right]\left(-\frac{\partial}{\partial x}\right)^n\right\}_{\text{even}}\right.$$

$$\left.+\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^{\infty}\frac{\mathrm{d}x}{2\pi}\left\{\operatorname{res}_h\left[\frac{\mathrm{e}^{\frac{\mathrm{i}hx}{p}}}{s+h}\sigma_1(h)\right]\bar{F}_0(x,p)^n\left(-\frac{\partial}{\partial x}\right)^n\right\}_{\text{even}}\right)\log\big(1+\mathrm{e}^{-2\pi x}\big).$$

We can shift the summation index in the first sum and then use (4.9) to calculate both sums. This gives us

(4.10)

$$\zeta_1(s,p) = \operatorname{res}_h\left[\sigma_1(h)\zeta_0(h,p)p^{-1}(1+s-h)\zeta_R(1+s-h,p)\right] + \operatorname{res}_h\left[\frac{\sigma_1(h)\zeta_R(-h)}{s+h}\right].$$

To bring the answer to the final form we can use the duality property of $S(h)$ which in the new variables simply reads

$$\begin{aligned} S(h)S(-h)=1 \quad &\Rightarrow \quad \sigma_1(h)+\sigma_1(-h)=0,\\ &\quad\ \sigma_1(h)\sigma_1(-h)+\sigma_2(h)+\sigma_2(-h)=0,\\ &\quad\ \dots. \end{aligned} \tag{4.11}$$

Then the last term in the formula (4.10) can be modified and added to the first one,

$$\begin{aligned} \zeta_1(s,p) &= \mathrm{res}_h\big[\sigma_1(h)\zeta_0(h,p)p^{-1}(1+s-h)\zeta_R(1+s-h,p)\big]+\mathrm{res}_h\left[\frac{\sigma_1(h)\zeta_R(h)}{s-h}\right]\\ &= \mathrm{res}_h\big[\sigma_1(h)\zeta_0(h,p)p^{-1}(1+s-h)\zeta_R(1+s-h,p)\big]+\mathrm{res}_h\left[\frac{\sigma_1(h)\zeta_0(h)}{s-h}\right]\\ &= \mathrm{res}_h\big[\sigma_1(h)\zeta_0(h,p)p^{-1}(1+s-h)\zeta_0(1+s-h,p)\big]. \end{aligned}$$

That is exactly the result (3.3) we obtain from the exponential representation.

To proceed with higher orders of the expansion of $\Psi_\nu(s,p)$, let us introduce a shorthand notation for the integral (4.1) in form of the "contraction"

$$\begin{aligned} &\langle g,f\rangle_{n'}\\ &\quad=-\frac{2}{p}\sum_{n=0}^{\infty}\frac{1}{n!}\int_0^\infty\frac{\mathrm{d}x}{2\pi}\left\{\mathrm{res}_h\left[\mathrm{e}^{\frac{\mathrm{i}hx}{p}}\frac{f(h)}{s+h}\bar F_0(x,p)^n f(x,p)\right]\left(-\frac{\partial}{\partial x}\right)^{n+n'}\right\}_{\text{even}}\log\big(1+\mathrm{e}^{-2\pi x}\big). \end{aligned}$$

Then the formula (4.9), for example, takes a simple form

$$\langle 1,1\rangle_{n'}=\begin{cases}(\mathrm{i}p)^{-n'}(1+s)_{n'}\zeta_R\big(s+n',p\big) & \text{for } n'=0,1,\\ (\mathrm{i}p)^{-n'}(1+s)_{n'}\zeta_0\big(s+n',p\big) & \text{for } n'\geq 2.\end{cases}$$

For $\zeta_1(s,p)$, we have

$$\zeta_1(s,p)=\big\langle 1,\bar F_1\big\rangle_1+\langle\sigma_1,1\rangle_0.$$

We apply (4.11) to the last term and add to the first one to replace $\zeta_R(1+s,p)$ by $\zeta_0(1+s,p)$ as it was done above.

Now let us discuss the general case $\zeta_k(s,p)$. We will prove Proposition 3.1 by induction. Suppose it is correct for some $k-1$. Using the "contraction" defined above, we write

$$\begin{aligned} \zeta_k(s,p) &= \big\langle 1,\bar F_k\big\rangle_1+\big\langle 1,\bar F_{k-1}\bar F_1\big\rangle_2+\big\langle 1,\bar F_{k-2}\bar F_2\big\rangle_2+\frac{1}{2!}\big\langle 1,\bar F_{k-2}\bar F_1^2\big\rangle+\cdots\\ &\quad+\langle\sigma_k(h),1\rangle_0+\big\langle\sigma_{k-1}(h),\bar F_1\big\rangle_1+\left(\big\langle\sigma_{k-2},\bar F_2\big\rangle_1+\frac{1}{2}\big\langle\sigma_{k-2},\bar F_1^2\big\rangle_2\right)+\cdots. \end{aligned} \tag{4.12}$$

The very first term is

$$\begin{aligned} &\big\langle 1,\bar F_k\big\rangle_1\\ &\quad=\mathrm{res}_h\big[(\sigma_1(h)\zeta_{k-1}(h)+\cdots+\sigma_{k-1}(h)\zeta_1(h,p))(1+s-h)p^{-1}\zeta_R(1+s-h,p)\big]. \end{aligned} \tag{4.13}$$

If it had the function $\zeta_0(1+s-h,p)$ instead of $\zeta_R(1+s-h,p)$, the first line would have given us the exponent (3.4), which is the direct consequence of the integral (4.9). It can be shown that the terms coming from the second line of (4.12) are exactly the missing singular terms we need to add to (4.13) to replace $\zeta_R(1+s-h)$ by

$\zeta_0(1+s-h,p)$. Indeed, let us apply the duality (4.11),

$$\begin{aligned}\langle\sigma_k,1\rangle_0 &= \operatorname{res}_h\left[\frac{\sigma_k(h)}{s+h}\zeta_0(-h,p)\right]\\ &= \operatorname{res}_h\left[\frac{\sigma_k(h)}{s-h}\zeta_0(h,p)\right]+\sum_{i=1}^{k-1}\operatorname{res}_h\left[\frac{\sigma_i(h)\sigma_{k-i}(-h)}{s-h}\zeta_0(h,p)\right].\end{aligned}\tag{4.14}$$

Now in (4.12) take the terms which are the "contractions" of $\sigma_i(h)$ with some powers of $\bar F_j$. Once again, we will get the function ζ_R in one of such "contractions"

$$\begin{aligned}\big\langle\sigma_i,\bar F_{k-i}\big\rangle_1 = \operatorname{res}_h\operatorname{res}_{h'}\Bigg[&\frac{\sigma_i(h)}{s+h}\big(\sigma_{k-i}\big(h'\big)\zeta_0(h',p)+\cdots+\sigma_1\big(h'\big)\zeta_{k-i-1}\big(h'\big)\big)\\ &\times\big(1-h-h'\big)p^{-1}\zeta_R\big(1-h-h',p\big)\Bigg].\end{aligned}$$

The missing singular term is exactly the term proportional to $\sigma_i(h)$ coming from (4.14). We write, therefore,

$$\begin{aligned}\big\langle\sigma_i,\bar F_{k-i}\big\rangle_1+\operatorname{res}_h\left[\frac{\sigma_i(h)\sigma_{k-i}(-h)}{s-h}\zeta_0(h,p)\right] = \operatorname{res}_h\operatorname{res}_{h'}\Bigg[&\frac{\sigma_i(h)}{s-h}\\ \times\big(\sigma_{k-i}\big(h'\big)\zeta_0\big(h',p\big)+\cdots+\sigma_1\big(h'\big)\zeta_{k-i-1}\big(h'\big)\big)&\big(1-h-h'\big)\frac{1}{p}\zeta_0\big(1-h-h',p\big)\Bigg].\end{aligned}$$

Now, by the assumption of induction all terms the terms which are a "contraction" of $\sigma_i(h)$ with some powers of $\bar F_j$ sum up into

$$\sum_{i=1}^{k-1}\operatorname{res}_h\left[\frac{\sigma_i(h)\zeta_{k-i}(h,p)}{s-h}\right],$$

and these are exactly the singular terms we need to be able to replace ζ_R by ζ_0 in (4.13). This observation finishes the proof of Proposition 3.1.